Disclaimer: This code is not up to par and does not fit most of the assignment requirements. I am having trouble understanding how to do it. This is my practice code.

```
#include<iostream>
#include<list>
#include<cstring>
using namespace std;

class HashTable
{
  private:
    static const int hashGroups = 10;
    list<pair<int, string>> table[hashGroups];

    public:
      bool isEmpty() const;
      int hashFunction(int key);
      void insertItem(int key, string value);
      void removeItem(int key);
      string searchTable(int key);
      void printTable();
};

bool HashTable::isEmpty() const
{
  int sum{};
  for(int i{}; i < hashGroups; i++)
  {
    sum += table[i].size();
  }

  if(!sum)
  {
    return true;
  }
  return false;
}

int HashTable::hashFunction(int key)
{
  return key%hashGroups;
}

void HashTable::insertItem(int key, string value)
```

```
{
  int hashValue = hashFunction(key);
  auto& cell = table[hashValue];
  auto bItr = begin(cell);
  bool keyExists = false;
  for(; bItr != end(cell); bItr++)
  {
    if(bItr -> first == key)
    {
      keyExists = true;
      bItr -> second = value;
      cout<<"[WARNING] Key exists. Value replaced." << endl;
      break;
    }
  }

  if(!keyExists)
  {
    cell.emplace_back(key, value);
  }
  return;
}

void HashTable::removeItem(int key)
{
  int hashValue = hashFunction(key);
  auto& cell = table[hashValue];
  auto bItr = begin(cell);
  bool keyExists = false;
  for(; bItr != end(cell); bItr++)
  {
    if(bItr -> first == key)
    {
      keyExists = true;
      bItr = cell.erase(bItr);
      cout<<"[INFO] Item Removed." << endl;
      break;
    }
  }

  if(!keyExists)
  {
    cout<<"[WARNING] Key not found. Pair not removed."<< endl;
  }
```

```
}

void HashTable::printTable()
{
  for(int i{}; i < hashGroups; i++)
  {
    if(table[i].size() == 0) continue;

    auto bItr = table[i].begin();
    for(; bItr!=table[i].end(); bItr++)
    {
      cout<< "[INFO] Key: " << bItr -> << "Value: " << bItr -> first << endl;
    }
  }
}

int main()
{
  int number = 7983;
  string Name = Tim;
  insertItem(7983, Tim);
}
```